

**Title: Supply Chain Optimisation using Linear Programming and
Mixed Integer Linear Programming**

[A Cost Minimisation and Carbon Tax Modelling Project using Excel Solver]

Executive Summary

This project applies optimisation modelling to support supply chain distribution decisions. A Linear Programming model is first developed to minimise total transportation cost while satisfying factory capacity, warehouse capacity, and retailer demand constraints. The model is then extended into a Mixed Integer Linear Programming framework by incorporating binary facility activation decisions, fixed operating costs, and carbon emission tax. The LP model identifies the lowest-cost distribution structure, while the MILP extension evaluates the trade-off between cost efficiency and environmental sustainability. The results show that route allocation, facility utilisation, and carbon-related costs significantly influence the final distribution strategy. Overall, the project demonstrates how optimisation modelling can support data-driven and sustainable supply chain decision-making

Table of Contents

Introduction	1
1. LP Model Problem Definition	1
1.1 Linear Programming (LP) Model	2
1.2 Objective Function	2
1.3 Constraints for the Model	3
1.4 Factory Capacity Constraints.....	4
1.5 Warehouse Capacity Constraints.....	4
1.6 Demand Constraints of Retailers	4
1.7 Implementation in Excel Solver	4
1.8 Optimal Distribution Strategy.....	5
1.8.1 Summary of the Model.....	5
1.8.2 Critical Evaluation of the Model	6
1.9 Sensitivity Analysis and Managerial Insights.....	6
2. Case Study (Assumed)	8
2.1 Mixed-Integer Linear Programming Model.....	9
2.2 Purpose of the Model and Learning Outcome	9
Objective Function	9
Constraints for the Model.....	9
Factory Capacity Constraints	9
Warehouse Capacity Constraints	10
Retailer Demand Constraints	10
Carbon Tax Constraints	10
Binary Constraints	10
Implementation of Solver	10
Summary and Analysis of the Model.....	11
Cost Behavior	11
Binary Facility Decisions	12
Sustainability and Strategic Value.....	12
Conclusion.....	12

Introduction

Optimisation modelling is widely used in supply chain and logistics decision-making to allocate resources efficiently, minimise distribution costs, and evaluate operational constraints. This project develops a Linear Programming model to identify the minimum-cost shipment plan across factories, warehouses, and retailers while satisfying supply, capacity, and demand constraints.

The model is then extended into a Mixed Integer Linear Programming model by adding binary facility activation decisions, fixed operating costs, and carbon emission costs. This allows the analysis to reflect both operational routing decisions and strategic sustainability considerations.

1. LP Model Problem Definition

The dataset represents a transportation problem involving factories, warehouses, retailers, and multiple products. Based on this dataset, a transportation optimisation model is developed. The dataset consists of 3 factories, 4 warehouses, 5 retailers, and 3 products.

Route: Factories to Warehouses					Total Factory Capacity		Units	Total customer demand		
	W1	W2	W3	W4			10000	D1	D2	D3
F1										
F2										
F3										
F4										
F5										
F6										
W1										
W2										
W3										
W4										

Figure 1: Dataset for LP Model

For each **Product P**, the following continuous decision variables are defined:

$P(F,W)$ = quantity of product P delivered from factory F to warehouse W

$P(F,R)$ = quantity of product P delivered from factory F to retailer R

$P(W,R)$ = quantity of product P delivered from warehouse W to retailer R

All decision variables are constrained to be non-negative, as shipment quantities cannot be less than zero.

1.1 Linear Programming (LP) Model

The Linear Programming (LP) model provides a powerful and flexible framework for analysing and supporting business decisions. In this project, the LP model is used to minimise transportation cost while satisfying factory capacity, warehouse capacity, flow-balance, and retailer demand constraints. The term linear refers to the mathematical structure of the objective function and constraints, where relationships are expressed using linear equations or inequalities.

1.2 Objective Function

The objective of the transportation model is to minimise total transportation cost across all factory-to-warehouse, factory-to-retailer, and warehouse-to-retailer shipment routes. Total transportation cost is calculated as the sum of shipment quantities multiplied by their corresponding route costs.

$$\text{Min } Z = \sum X_FWP \times C_FW + \sum X_FRP \times C_FR + \sum X_WRP \times C_WR$$

Where:

X_FWP = quantity of product P shipped from factory F to warehouse W

X_FRP = quantity of product P shipped from factory F to retailer R

X_WRP = quantity of product P shipped from warehouse W to retailer R

C_FW = unit transportation cost from factory F to warehouse W

C_FR = unit transportation cost from factory F to retailer R

C_WR = unit transportation cost from warehouse W to retailer R



Figure 2: Objective Function of LP Model

1.3 Constraints for the Model

The image shows the 'Solver Parameters' dialog box in Excel. The 'Set Objective:' field is empty. The 'To:' section has three radio buttons: 'Max' (unselected), 'Min' (selected), and 'Value Of:' (unselected). The 'Value Of:' field contains the number '0'. The 'By Changing Variable Cells:' field is empty. The 'Subject to the Constraints:' section contains a list of constraints, with one constraint highlighted in green. To the right of the list are buttons for 'Add', 'Change', 'Delete', 'Reset All', and 'Load/Save'. Below the list is a checkbox labeled 'Make Unconstrained Variables Non-Negative' which is checked. The 'Select a Solving Method:' dropdown is set to 'Simplex LP'. There is an 'Options' button next to it. At the bottom, there are buttons for 'Help', 'Solve', and 'Close'. A text box at the bottom explains the solving methods: 'Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.'

Figure 3: Constraints of the LP Model

1.4 Factory Capacity Constraints

From 3 factories 3 products will be delivered to 4 warehouses and 5 retailers. For each factory and each product, the total number of delivery or outflow (to all warehouses and retailers) $\leq$ its capacity.

For each of the factory F and the product P:

$$\sum P(F,W) + \sum P(F,R) \leq \text{Factory Capacity}$$

1.5 Warehouse Capacity Constraints

In this problem, 4 Warehouses receive products from factories and deliver to the 5 retailers as they receive, and each warehouse receives the products less than its capacity.

For each Warehouse and product P:

$$\sum P(F,W) = \sum P(W,R) \text{ and}$$

$$\sum P(F,W) \leq \text{Warehouse Capacity}$$

All warehouses can receive the sum of products less than or equal of their capacities and can outflow the number of products as they receive.

1.6 Demand Constraints of Retailers

Each retailer has specific product demand which must be fulfilled by direct factory transportation from factory and the delivery from warehouse.

$$\sum P(F,R) + \sum P(W,R) \geq \text{Demand}$$

1.7 Implementation in Excel Solver

- a) Layout of decisions variables in blocks: Firstly, re-arrange the tables product-wise and then add tables for each product,
 - 3×4 table for (factories-warehouses)
 - 3×5 table for (factories-retailers)
 - 4×5 table for (warehouses-retailers)
- b) **Set Objective Cell:** one SUMPRODUCT that multiplies each flow(delivery) block by its corresponding cost matrix and adds them.
- c) **Set Factory Capacity Constraints:** For each factory,
 $\sum P(F,W) + \sum P(F,R) \leq \sum F_Capacity$ means the sum of factory-warehouse + factory-retailer $\leq$ the capacity of a product for that factory.
- d) **Set Warehouse Capacity Constraints and Balance:**

- Capacity: sum of receives is $\leq$ capacity of warehouses
 - Balance: sum of inbound = sum of outbound
- e) **Set Demand Constraints:** sum of delivery (factory-retailer + warehouse-retailer) is = the demand of a retailer of each product.

f) Solver Setting:

- Set the objective cell to Min
- Changing Variables: the whole block of decision variables
- Add all capacities, flows, and demand constraints
- Select Simplex LP method

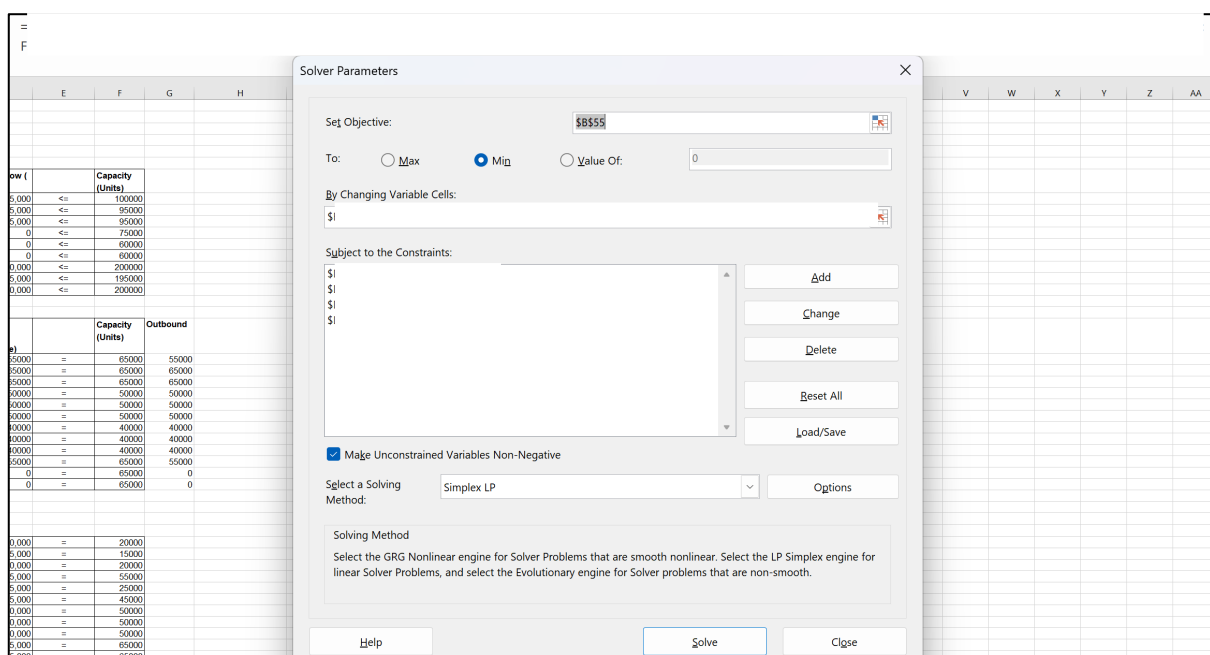


Figure 4 Solver parameter

1.8 Optimal Distribution Strategy

1.8.1 Summary of the Model

The minimum Total transportation cost is optimized by the modelling.

Factory 3 is the dominant supplier, but Factory 2 is not used at all. Although factory 1 plays a supportive role. The solution suggests that Factory 3 plays the dominant supply role, while Factory 1 provides supporting capacity and Factory 2 is not selected in the optimal LP solution.

1.8.2 Critical Evaluation of the Model

a) **Strengths:**

- The model captures all key points of the distribution problem.
- It shows a cost-minimizing solution.
- The model highlights economic insights- why it should emphasize factory 3.

b) **Limitations:**

- Single period, deterministic demand
- No fixed or handling cost

c) **Possible Extensions:**

- Add time-based capacity.
- Introduce multi-period planning.
- Include fixed cost for using each warehouse, so the model can also decide which warehouses to open or close.

1.9 Sensitivity Analysis and Managerial Insights

The sensitivity analysis evaluates the optimal transportation solution and identifies the parameters that have a strong influence on total cost.

- From the **variable cells section**, the shipment routes that carry a positive flow have a reduced cost of zero, confirming that these routes are part of the optimal basis. In contrast, several unused routes have positive reduced costs, indicating that they are not cost-effective. For example, factory 1 to warehouse 1.
- In the Constraints section, several **capacity constraints** are shown to be binding, as the values achieved in the optimal solution exactly match their respective limits. Some capacity constraints are binding, meaning the available capacity is fully utilised. The shadow price indicates how changes in capacity could affect total transportation cost within the allowable range.

	A	B	C	D	E	F	G	H
1	Microsoft Excel 16.0 Sensitivity Report							
2	Worksheet: [U 2573293.xlsx]Task 2							
3	Report Created: 12/12/2025 12:25:50 PM							
4								
5								
6	Variable Cells							
7								
8		Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
9		\$B\$5	Factory 1 Warehouse 1					
10		\$C\$5	Factory 1 Warehouse 2					
11		\$D\$5	Factory 1 Warehouse 3					
12		\$E\$5	Factory 1 Warehouse 4					
13		\$B\$6	Factory 2 Warehouse 1					
14		\$C\$6	Factory 2 Warehouse 2					
15		\$D\$6	Factory 2 Warehouse 3					
16		\$E\$6	Factory 2 Warehouse 4					
17		\$B\$7	Factory 3 Warehouse 1					
18		\$C\$7	Factory 3 Warehouse 2					
19		\$D\$7	Factory 3 Warehouse 3					
20		\$E\$7	Factory 3 Warehouse 4					
21		\$B\$10	Factory 1 Retailer 1					
22		\$C\$10	Factory 1 Retailer 2					
23		\$D\$10	Factory 1 Retailer 3					
24		\$E\$10	Factory 1 Retailer 4					
25		\$F\$10	Factory 1 Retailer 5					
26		\$B\$11	Factory 2 Retailer 1					
27		\$C\$11	Factory 2 Retailer 2					
28		\$D\$11	Factory 2 Retailer 3					
29		\$E\$11	Factory 2 Retailer 4					
30		\$F\$11	Factory 2 Retailer 5					
31		\$B\$12	Factory 3 Retailer 1					
32		\$C\$12	Factory 3 Retailer 2					
33		\$D\$12	Factory 3 Retailer 3					
34		\$E\$12	Factory 3 Retailer 4					
35		\$F\$12	Factory 3 Retailer 5					
36		\$B\$15	Warehouse 1 Retailer 1					
37		\$C\$15	Warehouse 1 Retailer 2					
38		\$D\$15	Warehouse 1 Retailer 3					
39		\$E\$15	Warehouse 1 Retailer 4					
40		\$F\$15	Warehouse 1 Retailer 5					
41		\$B\$16	Warehouse 2 Retailer 1					
42		\$C\$16	Warehouse 2 Retailer 2					
43		\$D\$16	Warehouse 2 Retailer 3					

Figure 5: Sensitivity Analysis

2. Case Study (Assumed)

Using the transportation cost table, develop a MILP formulation based on this cost table, and fixed costs to calculate the optimal delivered quantities that minimize the cost of total distribution for both products while mitigating all the demand and supply constraints.

Using the carbon-emission cost information, extend the model by incorporating environmental penalty values and evaluate how the recommended shipment routes change. Demand is higher in Period 2 than in Period 1, allowing the model to evaluate how cost and carbon impact change when demand increases.

[illegible]

Figure 6: Case study data for Transportation with Carbon Tax (Assumed)

2.1 Mixed-Integer Linear Programming Model

Mixed-Integer Linear Programming (MILP) is an optimisation technique that extends linear programming by allowing both continuous and integer decision variables within the same model. In this project, continuous variables represent shipment quantities, while binary variables represent whether a factory or warehouse is open or closed.

MILP is suitable for this supply chain problem because the model needs to optimise delivery quantities while also making facility activation decisions. This makes it useful for analysing transportation planning, warehouse selection, production allocation, fixed operating costs, and carbon emission costs within one structured framework.

2.2 Purpose of the Model and Learning Outcome

This MILP Model is developed to optimize a supply chain network consisting of factories, warehouses, and retailers for two planning periods. The primary objective is to minimize transportation cost, fulfill the demand and explicitly incorporate environmental sustainability through Carbon Emission Tax.

The use of MILP is fully justified because the problem involves both the continuous decisions (Transportation quantities), and the discrete strategic decisions (decision to open the unit or to close). Linear Programming alone is not sufficient for these realities, whereas MILP optimization in logistics network design is well established in supply chain modelling. Complex distribution environments require simultaneous operational decisions about routing together with strategic decisions such as facility activation, making MILP an appropriate tool for these joint decisions.

Objective Function

The objective function of this green transportation model is to minimize the cost of total transportation for 2 periods (years) by fulfilling the retailers' demands.

So, the Minimum Cost,

$$\text{Min } Z = \sum P1 (F-W) + \sum P1 (W-R) + \sum P2 (F-W) + \sum P2 (W-R) + \sum P1 (\text{Carbon Tax}) + \sum P2 (\text{Carbon Tax}) + \sum P1 (\text{Fixed Cost}) + \sum P2 (\text{Fixed Cost})$$

[Where, P1= Period 1, P2= Period 2, F= Factory, W= warehouse, and R= Retailer]

Constraints for the Model

Factory Capacity Constraints

From 2 factories, 2 products will be delivered to 2 warehouses and 3 retailers. For each factory and each product, the total number of deliveries or outflow (to all warehouses and retailers) $\leq$ its capacity.

Warehouse Capacity Constraints

In this problem, 2 Warehouses receive products from factories and deliver to the 3 retailers as their capacities. So all warehouses can receive the sum of products less than or equal to their capacities and can outflow $\leq$ the number of products as they receive.

Retailer Demand Constraints

For each retailer and product demand must be fully met by warehouse delivery.

Carbon Tax Constraints

- **Factory to Warehouse Carbon Tax Constraints,**
- **Warehouse to Retailer Carbon Tax Constraints**

Binary Constraints

If the facility (factory or warehouse) is open in the period, then 1; otherwise 0. These binary variables activate fixed operating costs and real capacity for the facility.

Implementation of Solver

The Total cost of transportation plus carbon emission tax is set as the objective function for this green transportation model. The supply unit from factory to warehouse, factory to retailer, and warehouse to retailer are set as the changing variable of the model. After that, 6 constraints are set for the model to minimize cost, and the model is run.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Buttons: Add, Change, Delete, Reset All, Load/Save, Options, Help, Solve, Close

Figure 7: Solver Parameter for MILP

Summary and Analysis of the Model

The MILP model produces a higher total cost than the basic LP model because it includes fixed facility costs and carbon emission penalties. The results show how facility activation, delivery quantities, and carbon-related costs affect the final distribution strategy.

	Transportation Cost					Transportation Cost					
	Carbon Emission Cost					Carbon Emission Cost					
	Total					Total					
Period 1						Period 2					
Binary variables	Delivery	Capacity	Open/Close	Fixed Cost	Real Capacity	Binary variables	Delivery	Capacity	Open/Close	Fixed Cost	Real Capacity
Factory 1											
Factory 2											
Warehouse 1											
Warehouse 2											
Total fixed cost				108000		Total Fixed Cost				108000	
Grand Total											

Figure 8: Summary of the Model

The figure below shows the total number of units delivered from each facility for the two periods.

	A	B	C	D	E	F	G	H	I	J	K
1	Period 1						Period 2				
2	Product 1						Product 1				
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											
18											
19											
20											
21											
22											

Figure 9: Number of Deliveries from each Facility

Cost Behavior

The total cost increases from Period 1 to Period 2 as demand rises. Carbon emission cost also increases proportionally, showing how demand growth affects both financial and environmental performance.

Binary Facility Decisions

All facilities remain open in the model. Closing of any facility will increase transportation distance, and emissions. This is a strong demonstration of why MILP is superior to LP for strategic supply chain design.

Sustainability and Strategic Value

The explicit pricing of carbon emissions transforms sustainability into a quantifiable decision maker rather than a qualitative objective. The model discourages inefficient routing and shows how demand growth increases environmental costs unless strategic facilities are adopted. This directly aligns with ESG-driven supply chain planning and carbon taxation frameworks.

Conclusion

This project demonstrates how optimisation modelling can support supply chain distribution decisions. The Linear Programming model provides a structured approach for minimising transportation cost while satisfying factory capacity, warehouse capacity, and retailer demand constraints. The Mixed Integer Linear Programming extension improves the model by incorporating facility activation decisions, fixed operating costs, and carbon emission costs. Overall, the project shows how Excel Solver can be used to evaluate cost-efficient and sustainability-aware supply chain strategies.